

창의기계설계 (BE203-01)

학기	2026년 1학기
담당교수	박준혁
이수구분	기초필수
학점	3.0
강의시간	금09:00-12:00(E7 - 307)
강의실	E6 211 (교수사무실)
소속	기초학부

교과개요

본 교과목은 기계공학을 비롯한 다양한 공학 분야에서 필수적으로 요구되는 부품 및 시스템의 설계·제작 개념을 학습하고, 이를 구현하기 위한 기술적 방법과 실습 중심의 접근법을 다룬다. 학생들은 2D 및 3D CAD 설계를 통해 유요소자 및 강성기계 시스템을 디자인하고, 유한요소해석 기법(Finite Element Analysis, FEA)을 통해 시스템을 모델링하며, Autodesk TinkerCAD를 이용한 전자회로 및 기본적인 Arduino 기반 회로 설계 및 관련 코딩을 실습하여 다양한 전기기계적 시스템에 대한 설계와 전자회로와의 제어 및 융합을 경험한다. 최종적으로 팀 단위 메카트로닉 시스템 설계를 수행하여, 우수한 제안서에 있어서는 교원의 지도 하에 UGRP를 통한 설계 경진대회 참가, 스타트업, 논문 제출 등으로 아이디어를 발전시킬 수 있는 기회 및 이론적, 기술적 소양을 쌓는 것을 목표로 한다.

This course covers the fundamental concepts of design and fabrication of components and systems required across mechanical engineering and related disciplines, emphasizing both technical methodologies and hands-on practice. Students will model mechanical elements (both in soft and rigid systems) through 2D and 3D CAD design, and gain practical experience in electronic circuit design and basic Arduino-based circuit programming using Autodesk TinkerCAD—thereby integrating soft/rigid electro-mechanical design with electronic circuits and control. Ultimately, students will develop the theoretical and technical competencies to draft a team-based mechatronic system design proposal with a potential option to be continued further as a Undergraduate Group Research Projects (UGRP) for design competitions, entrepreneurship applications, or scientific publications under the instructor's supervision.

교과목표

기계 및 융합공학 분야에서 설계·제작의 기초 개념과 프로세스를 이해한다.

Understand the fundamental principles and processes of design and manufacturing in mechanical and multidisciplinary engineering.

2D/3D CAD 디자인 기술을 습득하고 부품 및 조립체를 설계한다.

Acquire proficiency in 2D/3D CAD modeling and assembly design.

(Rigid Mechanics) 기계요소(볼트, 너트, 베어링, 기어 등) 및 동력전달 장치(기어, 캠, 벨트 등)의 설계원리를 학습한다.

Learn the design principles of mechanical elements (bolts, nuts, bearings) and power transmission components (gears, cams, belts).

TinkerCAD를 활용한 전자회로 및 아두이노 기초 프로그래밍을 실습한다.

Practice basic electronics and Arduino programming using Autodesk TinkerCAD.

창의적프로젝트를 통해 아이디어를 구상 및 설계한다.

Apply design thinking through a creative, team-based design project.

학습윤리

Students should familiarize themselves with the general rules of academic conduct, and students are prohibited from engaging in

- Unauthorized assistance: Using or attempting to use unauthorized materials, information or study aids in any academic exercise unless specifically authorized by the instructor of record. The unauthorized possession of examination or course-related material also constitutes cheating.
- Communication to another through written, visual, electronic, or oral means: The presentation of material which has not been studied or learned, but rather was obtained through someone else's efforts and used as part of an examination, course assignment, or project.
- Commercial Use of Academic Material: Selling of course material to another person, student, and/or uploading course material to a third-party vendor without authorization or without the express written permission of the university and the instructor. Course materials include but are not limited to class notes, Instructor's PowerPoints, course syllabi, tests, quizzes, labs, instruction sheets, homework, study guides, handouts, etc.
- Falsifying or misrepresenting the student's own academic work, especially using AI (e.g., ChatGPT). If you used AI for any part of your assignment, make sure you indicate which part of your assignment was done using AI.
- Plagiarism: Using or appropriating another's work without any indication of the source, thereby attempting to convey the impression that such work is the student's own.
- Multiple Submissions: Submitting the same academic work from your past course(s) or from other course(s) you are taking in the same semester without the express written permission of the instructor.
- Soliciting assistance with academic coursework and/or degree requirements.

교과정책(방침)

Using mobile devices for anything other than course-related search is prohibited. If you need to respond to any text messages or calls that cannot wait, do it quickly without disturbing others in class or lab sessions.

No recording or taking photos of any kind during lectures or labs

준비물 및 기타

Required Text: None

Suggested Texts:

- o Shigley's Mechanical Engineering Design, 11th Ed., McGraw-Hill
- o Engineering Design Graphics with AutoCAD, J.D. Bethune, Pearson
- * Supplementary lecture/lab materials will be provided

Required Hardware:

- o Laptop for assignments

Required Software:

- o Autodesk AutoCAD
- o Autodesk Fusion 360
- o Autodesk TinkerCAD Tutorial
(<https://www.tinkercad.com/>)
- o Arduino IDE (Arduino.cc)